SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application BasedQuestions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Maddipati Teja	**Reg. No.:**	192425323
Section / Batch:	DAA, 2024	**Date:**	21-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:		**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

de of Conduct

M. Tesce (Name / Reg No) certify that this submission is my original work and that I have dhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will esult in disciplinary action.

Signature of the Student: M. Tesce

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______	Signature: ______	Signature: ______
Date: ______	Date: ______	Date: ______

Application: online video streaming platform

A video streaming processess user data using an algorithm with recurrence

$$[T(n) = 2T(n/2) + n]$$

Applying master theorem

$$T(n) = 2T(n/2) + n$$

Comparing with master theorem

$$T(n) = aT(n/b) + f(n)$$

$a = 2$
$b = 2$
$f(n) = n$

1. $n^{\log_2 2} = n^1 = n$

2. $n^{\log_b a} = n$

2. $f(n)$

$$f(n) = n = n^{\log_b a}$$

$$f(n) = \Theta(n^{\log_b a})$$

Apply M.T

$$T(n) = \Theta(n^{\log_b a} \log n)$$

$$T(n) = \Theta(n \log n)$$

Final complexity

$$[\Theta(n \log n)]$$

Scalability Analysis.

↳ Data is divided Equally into two p

↳ processing grows almost linearly.

Asymptotic interpretation.

worst case : $O(n\log n)$

Best case : $\Omega(n\log n)$

Tight Bound : $\Theta(n\log n)$.

The algorithm is efficient for large scale video streaming systems.

Q2

Social media feed processing

A social media platform processes posts using

1 $T(n) = T(n-1) + n$

2 $T(n) = T(n-2) + (n-1) + n$

3 $T(n) = T(n-3) + (n-2) + (n-1) + n$

$T(n) = T(n-k) + \sum_{i=n-k+1}^{n} i$

$k = n-1$

$T(n) = T(1) + 2 + 3 + \dots + n$

$1 + 2 + \dots + n = \frac{n(n+1)}{2}$

$T(n) = O(n^2)$

Final complexity

$\left[\Theta(n^2)\right]$

SIMATS ENGINEERING (SAVEETH...
Department of Co...
ASSESSMENT ...

performance Analysis

↳ every recursive call perform additional work.

↳ Total work increase quadratically

↳ memory and processing time increases rapidly.

Asymptotic interpretation

worst case $O(n^2)$

Best case: $\Omega(n^2)$

Tight Bound: $\Theta(n^2)$

The algorithm is not suitable for very large scale social media datasets.

Q8

E-commerce Recommendation System

$\Theta(n\log n)$

$O(n^2)$

complexity $O(n\log n)$

Growth rate $n\log n$

Example

for $n=1000$

$1000\times\log_2(1000)\approx1000\times10=10000$

complexity $O(n^2)$

Growth Rate n^2

For

$n=1000$

$1000^2 = 1,000,000$

comparision table

Data size	$O(n \log n)$	$O(n^2)$
100	664	10,000
1,000	9,966	1,000,000
10,000	132,877	100,000,000

The $O(n \log n)$ algorithm is more scalable and efficient than $O(n^2)$ for e-commerce recommendation systems.

Cloud Storage processing System

$T(n) = 3T(n/2) + n$

$T(n) = 3T(n/2) + n$

$a = 3$

$b = 2$

$f(n) = n$

$n^{\log_2 3}$

$\log_2 3 \approx 1.585$

$n^{\log_2 3} = n^{1.585}$

$f(n) = n$

$n < n^{1.585}$

$f(n) = O(n^{\log_2 3 - \varepsilon})$

$(\varepsilon > 0)$

Hence master theorem case 1 applies.

$T(n) = \Theta(n^{\log_2 3})$

$\{\Theta(n^{1.585})\}$

worst case $O(n^{1.585})$

Best case $\Omega(n^{1.585})$

Tight Bound $\Theta(n^{1.585})$

VEETHA SCHOOL OF ENGINEERING)
nputer Science and Engineering
– ANALYTICAL PROBLEM SOLVING

Design and Analysis of Algorithms
Assessment Tool 1 – Analytical
Solving

Q5

Cinankial Analytics System

* Cinankial System uses

$T(n) = 2T(n/2) + n\log n$

$T(n) = 2T(n/2) + n\log n$

$a = 2$

$b = 2$

$f(n) = n\log n$

$n^{\log_2 2} = n$

$f(n) = n\log n$

$f(n) = \Theta(n^{\log_b a} \log^1 n)$

$T(n) = \Theta(n^{\log_b a} \log^{1+1} n)$

$T(n) = \Theta(n\log^2 n)$

$\Theta(n\log^2 n)$

Asymptotic interpretation

worst case $O(n\log^2 n)$

Best case $\Omega(n\log^2 n)$

Tight Bound, $\Theta(n\log^2 n)$

$T(n) = \Theta(f(n))$